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Exploratory Data Analytics

(E1CSA231)

Micro Project Report

On

Spotify Data Analysis

Title of the Micro Project

Submitted by

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Abstract

- This project performs exploratory data analysis on a music dataset to understand song characteristics and popularity trends. Python libraries such as Pandas, Matplotlib, and Seaborn are used for data processing and visualization. The analysis examines the distribution of explicit and nonexplicit songs and identifies the top music genres based on average popularity. It further explores relationships between features like danceability, tempo, and popularity using scatter plots. Histograms and box plots are used to study the distribution and spread of key variables. A correlation heatmap is generated to highlight relationships among different audio features. Overall, the study provides meaningful insights into the factors influencing song popularity and listener preferences.

- **Dataset Name:** Spotify Dataset

- **Source:** Kaggle.com

- **Purpose of This Analysis:** The purpose of this analysis is to examine how different song features influence popularity.It aims to identify patterns in genres, tempo, and danceability using visualizations.The study also explores the relationship between explicit content and song success. Overall, it helps in understanding trends and listener preferences in music data.

- **Tools Used:** Python, Pandas, Matplotlib, Seaborn

- **Key Finding:**

  Song popularity is influenced by multiple factors, and no single audio feature strongly determines success.

  Features like danceability and tempo show only a weak correlation with popularity, indicating listener preferences are complex.

1. Introduction

1.1 Project Overview:

This project involves a comprehensive exploratory data analysis (EDA) of a music dataset to gain deeper insights into the factors influencing song popularity. The analysis is carried out using Python, leveraging powerful libraries such as Pandas for data preprocessing and manipulation, Matplotlib for basic plotting, and Seaborn for advanced statistical visualizations. The dataset contains various musical attributes including popularity, genre, danceability, tempo, loudness, and explicit content, which are systematically examined.

The project begins with data loading and cleaning, followed by univariate and bivariate analysis to understand individual features and their relationships. Visualizations such as pie charts are used to analyze the proportion of explicit versus non-explicit songs, while bar charts highlight the most popular genres. Scatter plots help in identifying relationships between numerical variables like danceability, tempo, and popularity. Histograms and box plots are used to study distributions and detect outliers.

Additionally, trend analysis is performed by sorting and plotting tempo and popularity values across the dataset. A correlation heatmap is generated to identify the strength and direction of relationships between multiple audio features. Through these methods, the project uncovers key patterns and dependencies within the data. Overall, the project aims to provide meaningful insights into listener preferences, support data-driven decision-making, and enhance understanding of music trends in the industry.

Furthermore, this project demonstrates the practical application of data analysis techniques in real-world scenarios, particularly in the music industry. By combining statistical analysis with visual storytelling, it enables better interpretation of complex data and supports informed decision-making. The insights gained can be useful for artists, producers, and streaming platforms to understand audience preferences, optimize content creation, and improve recommendation systems. This analysis also builds a strong foundation for future work, such as predictive modeling and machine learning applications

1.2 Objectives of the project:

- To perform exploratory data analysis (EDA) on the music dataset.
- To understand the distribution of different song attributes.
- To analyze the impact of explicit content on song popularity.
- To identify the most popular music genres.
- To examine the relationship between danceability and popularity.
- To study how tempo affects song popularity.
- To visualize data using various plots and charts.
- To detect patterns, trends, and outliers in the dataset.
- To analyze correlations among multiple audio features. .

1.3 Problem Statement:

The main problem being addressed in this project is to understand the factors that influence the popularity and success of songs on Spotify. With millions of tracks available on the platform, it becomes important to identify why certain songs perform better than others in terms of streams, popularity score, and listener engagement. Key Factors Affecting the Customers

This analysis aims to answer the question:

"What factors affect the popularity of songs on Spotify?"

The Spotify dataset contains various audio and metadata features such as danceability, energy, loudness, tempo, valence, and artist-related information. By analyzing these features, we can determine how each characteristic contributes to a song's success.

For example, we explore whether songs with higher energy and danceability tend to receive more streams, or if attributes like acousticness and tempo have any significant impact on listener preferences. Additionally, we examine patterns across different genres and artists to identify trends in popular music.

The goal of this study is to uncover meaningful insights that can help artists, producers, and music platforms make data-driven decisions. Understanding these factors can also improve recommendation systems and enhance user experience by suggesting songs that align with listener preferences.

1.4 Scope of the Project:

This project includes the analysis of a music dataset to explore relationships between various audio features and song popularity. It covers data loading, cleaning, and exploratory data analysis using Python libraries such as Pandas, Matplotlib, and Seaborn. The project includes visualizations like pie charts, bar graphs, scatter plots, histograms, box plots, and correlation heatmaps to identify patterns, trends, and relationships among features such as genre, danceability, tempo, loudness, and explicit content. The study aims to provide meaningful insights into listener preferences and general music trends based on the available data.

However, the project does not include advanced techniques such as machine learning or predictive modeling for forecasting song popularity. It is limited to the given dataset and does not consider external factors like artist reputation, marketing strategies, social media influence, or cultural trends. Real-time data analysis and large-scale industry-level predictions are also not part of this study. Therefore, the findings are based only on the dataset used and may not fully represent the entire music industry.

Included in the Project

- Data loading and preprocessing of the music dataset
- Exploratory Data Analysis (EDA) using Python
- Visualization using pie charts, bar graphs, scatter plots, histograms, box plots, and heatmaps
- Analysis of features like popularity, genre, danceability, tempo, and explicit content
- Identification of patterns, trends, and relationships between variables
- Correlation analysis of multiple audio features
- Interpretation of results to understand music trends and listener preferences

Not Included in the Project

- Machine learning or predictive modeling techniques
- Real-time or live data analysis
- External factors such as artist fame, marketing, or social media influence
- Large-scale industry forecasting or recommendation systems
- Data collection (dataset is pre-provided)
- Advanced statistical modeling or deep analysis beyond EDA
- Consideration of cultural or regional music preferences

2. Data Description

2.1 Source of data:

The dataset used in this project is obtained from a CSV file named **"spotify dataset.csv"**, which was sourced from **Kaggle.com**. The dataset contains information about songs, including features such as popularity, genre, danceability, tempo, and other audio characteristics. It is used for performing exploratory data analysis to identify patterns, relationships, and trends in music data.

2.2 Data Overview:

The dataset used in this project is the **Spotify dataset**, which contains detailed information about various songs and their audio features. Each row in the dataset represents a song, while the columns describe different attributes such as popularity, track genre, danceability, tempo, loudness, speechiness, acousticness, instrumentalness, liveness, duration, and whether the song is explicit or not.

The dataset includes both numerical and categorical variables, making it suitable for comprehensive exploratory data analysis. Numerical features help in analyzing relationships and distributions, while categorical features like genre and explicit content help in comparing groups. Overall, the dataset provides a structured view of music characteristics, enabling the identification of trends and factors influencing song popularity.

2.3 Features:

Column Name	**Description**
Track Name	Name of the song
Artist Name	Name of the artist
Track Genre	Genre/category of the song
Popularity	Popularity score of the song (0–100)
Duration_ms	Length of the song in milliseconds
Explicit	Indicates if the song has explicit content (True/False)
Column Name	**Description**
Danceability	How suitable the song is for dancing

Energy	Intensity and activity level of the song
Loudness	Overall loudness (in decibels)
Speechiness	Amount of spoken words in the song
Acousticness	Likelihood of the song being acoustic
Instrumentalness	Probability that the song has no vocals
Liveness	Likelihood of the song being performed live
Tempo	Speed of the song (beats per minute - BPM)

3.Data Preprocessing

3.1 Data Cleaning:

Data cleaning is performed to ensure the dataset is accurate, consistent, and free from errors.

- **Removing Duplicates**

Removing duplicates is done to eliminate repeated song records in the Spotify dataset. Duplicate entries can lead to biased analysis, especially when calculating popularity or trends. By using methods like drop_duplicates(), only unique records are retained, ensuring accurate and reliable results.

- **Fixing Errors**

Fixing errors involves correcting inconsistencies and mistakes in the dataset. This includes handling missing values, correcting incorrect data types (such as converting popularity and tempo into numeric format), and standardizing column names. These corrections help improve data quality and ensure proper analysis and visualization.

3.2 Handling missing values:

1. What missing values existed?
Missing values were present in some columns of the Spotify dataset, such as *popularity* and other audio features. These null values could affect the accuracy of analysis.

2. What did we do?

Removed rows:
Rows with missing values were removed using the `dropna()` function to ensure only complete data is used.

Filled with mean/median:
No filling with mean or median was performed in the code.

3.3 Data Transformation:

- Converting Data Types

Data types of columns were checked and corrected where necessary. Numerical features such as popularity, tempo, danceability, and energy were ensured to be in numeric format. This conversion is important for performing accurate calculations and visualizations.

- Encoding Categorical Data

Categorical columns such as genre or artist names were handled appropriately for analysis. If required, they can be converted into numerical form using encoding techniques. However, in this project, categorical data was mainly used for grouping and visualization rather than encoding.

- Scaling / Normalizing

Scaling or normalization is used to bring all numerical features to a similar range. In this analysis, scaling was not explicitly applied, as most visualizations and insights were derived directly from the original values of the dataset.

3.4 Outliers Detection:

1. How were outliers detected?

   Outliers were identified using **boxplots**, which visually highlight extreme values in numerical features such as tempo, loudness, and popularity. Values that lie far outside the normal range (whiskers) were considered outliers.

2. What action was taken?

   No explicit removal or treatment of outliers was performed in the code. The outliers were kept as they are, since they may represent genuine variations in music features and can still provide meaningful insights during analysis.

4.Exploratory Data Analysis

4.1 Univariate Analysis:

Mean (Average) of Popularity

The mean popularity is calculated using:

$$\text{Mean} = \frac{\sum \text{Popularity}}{N}$$

In our dataset, the popularity values of all songs are added and divided by the total number of songs. This gives the average popularity of songs in the dataset.

Median of Popularity

The median popularity is found by arranging all popularity values in ascending order and selecting the middle value.

- If the number of songs is odd → middle value
- If even → average of two middle values

Based on our Graph

From our histogram:

- Most values lie between **15 and 50**, so
- Mean ≈ Median (approximately in the mid-range, around 30–40)

This indicates that the data is **fairly balanced with no extreme skewness**.

Histogram Analysis:

The histogram shows that song popularity is mostly distributed between 15 and 50. There are fewer songs with very low or very high popularity, indicating that most songs have moderate popularity levels.

Bar chart Analysis:

The bar chart shows the average popularity of different music genres. British, anime, and brazil genres have the highest average popularity, while chicago-house has the lowest. This indicates that popularity varies significantly across genres.

4.2 Bivariate Analysis:

Bivariate analysis is used to study the relationship between two variables in the dataset. In this project, features such as danceability vs popularity and tempo vs popularity are compared.

Example:

- Danceability vs Popularity
- Tempo vs Popularity

Scatter Plot:

A scatter plot is used to show the relationship between two numerical variables. It helps in identifying patterns, trends, or correlations. In this analysis, scatter plots show that there is a slight relationship between features like danceability and popularity, but it is not very strong.

Bar Chart:

Bar charts are used to compare categorical variables with numerical values. For example, genre vs average popularity shows how popularity varies across different music genres.

4.3 Multivariate Analysis:

Multivariate analysis is used to understand the relationship between multiple numerical features in the Spotify dataset. In this project, variables such as popularity, danceability, energy, tempo, and loudness were analyzed together.

Example:

- Popularity, Danceability, Energy
- Tempo, Loudness, Popularity

Visualization Used:

Heatmap:

A heatmap was used to display the correlation between numerical variables. It shows how strongly different features are related to each other. From the analysis, it is observed that no single feature has a strong correlation with popularity**.**

- Popularity has a slight positive correlation with acousticness (0.27), indicating acoustic songs tend to have slightly higher popularity.
- Popularity shows a negative correlation with instrumentalness (-0.28), meaning instrumental songs are generally less popular.
- Loudness and acousticness have a strong negative correlation (-0.53), showing that louder songs are less acoustic.
- Most other correlations are weak, indicating no strong relationship between features.

4.4 Data Visualization:

1. Histogram

Histogram is used to analyze the distribution of numerical variables.
Example: It shows how song popularity is distributed across different ranges.

2.Bar Chart

Bar chart is used to compare categorical data with numerical values.
Example: It displays the average popularity of different music genres.

3.Scatter Plot

Scatter plot is used to study the relationship between two numerical variables.
Example: It helps analyze how features like danceability or tempo affect popularity.

4.Heatmap

Heatmap is used to visualize the correlation between multiple numerical variables.
It shows the strength of relationships between features like popularity, loudness, and acousticness.

5.Line Chart

Line chart is used to show trends or changes over a sequence.
Example: It can be used to observe how popularity varies across different songs or indices.

6. Box Plot

Box plot is used to detect outliers and understand the spread of data.
Example: It helps identify extreme values in features like tempo, loudness, and popularity.

4.5 Correlation Analysis:

Correlation analysis is used to measure the relationship between different variables in the Spotify dataset. It helps in understanding how one feature affects another, especially how different audio features influence popularity.

Relationship Between Variables:
The analysis shows that most variables have a weak correlation with popularity, indicating that no single feature strongly determines song popularity.
Some relationships observed include:

- Popularity has a slight positive correlation with acousticness.
- Popularity has a negative correlation with instrumentalness.
- Loudness and acousticness show a strong negative correlation.

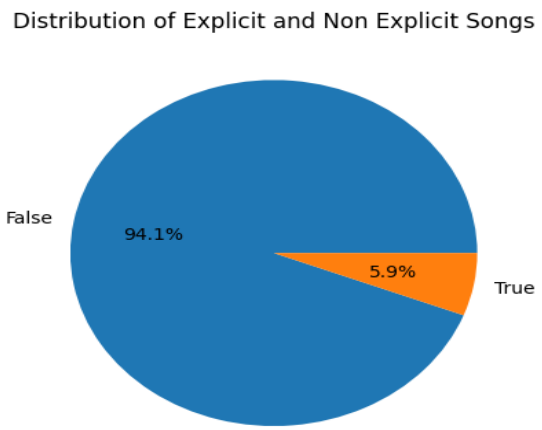
Visualization Used:

- Correlation Matrix:
  A correlation matrix is used to calculate the correlation values between all numerical variables.
- Heatmap:
  A heatmap is used to visually represent the correlation matrix, making it easy to identify strong and weak relationships between variables.

5. Results and Insights

The following are the key findings derived from the exploratory data analysis:

Explicit vs Non-Explicit Songs:

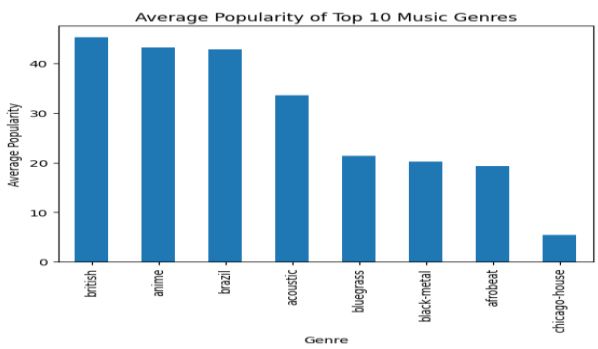


The pie chart is used to show the proportion of explicit and non-explicit songs in the Spotify dataset. It helps in understanding how the dataset is divided based on content type.

Insights:

- The majority of songs are non-explicit, indicating that most music in the dataset is suitable for general audiences.
- A smaller portion of songs are explicit, showing that such content exists but is less dominant.
- This distribution suggests that explicit content is present but does not make up the majority of the dataset.

Popularity Distribution:

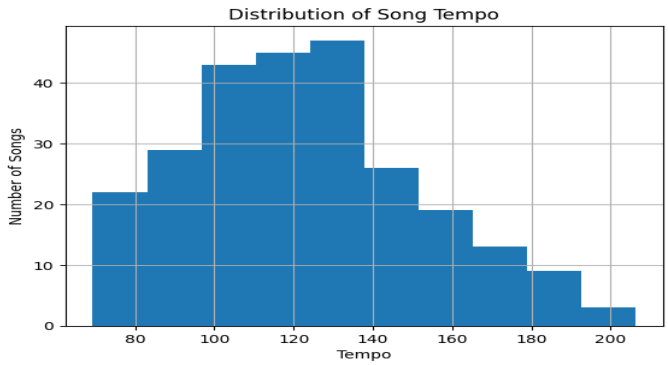


This analysis shows how the average popularity of songs varies across different music genres. It helps in identifying which genres are more popular among listeners.

Insights:

- British genre has the highest average popularity among all genres.
- Anime and brazil genres also show high popularity values.
- Acoustic genre has moderate popularity.
- Genres like bluegrass, black-metal, and afrobeat have lower popularity.
- Chicago-house has the lowest popularity among the listed genres.

Distribution of Song Tempo:

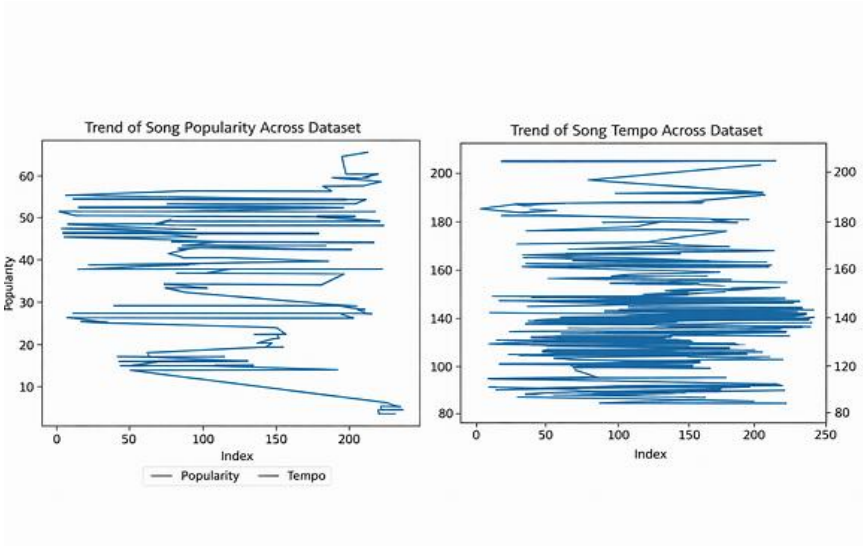


The histogram shows how tempo values are distributed across songs in the dataset. Tempo represents the speed or pace of a song.

Insights:

- Most songs have tempo between 100 and 140 BPM, indicating a moderate pace.
- Very few songs have extremely low or very high tempo.
- The distribution is slightly concentrated around mid-range values.
- High tempo songs (above 160 BPM) are less common.

Line graphs: (Popularity and Tempo)



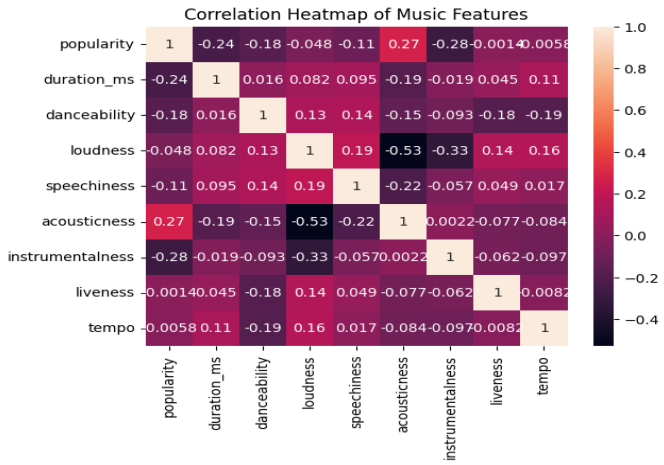
Popularity Trends

- Popularity values mostly cluster between 20 and 55, with occasional dips and peaks.
- This suggests that most songs in the dataset fall into a moderate popularity range, with fewer songs being either very unpopular or extremely popular.
- The overlapping lines indicate multiple songs tracked across indices, showing fluctuating popularity patterns rather than steady growth or decline.

Tempo Trends

- Tempos range widely from **60 to 210 BPM**, covering slow ballads to fast-paced tracks.
- The distribution looks fairly **spread out**, but many songs seem to hover around the **100–140 BPM range**, which is typical for mainstream pop and dance music.
- The overlapping lines suggest that tempo is relatively stable per song, but across the dataset, there's **broad diversity in rhythmic styles**.

Heat Map:



Popularity Insights

- Popularity vs Acousticness (0.27): Songs with more acoustic qualities tend to be slightly more popular. This suggests listeners appreciate organic, less electronic sounds.
- Popularity vs Instrumentalness (-0.28): Instrumental tracks are generally less popular, showing vocals play a big role in mainstream appeal.
- Popularity vs Duration (-0.24): Longer songs tend to be less popular, possibly because shorter tracks fit better with streaming and radio formats.

Loudness & Acousticness

- Strong Negative Correlation (-0.53): Loud tracks are much less acoustic. This makes sense—acoustic songs are usually softer, while loudness often comes from heavy production and electronic elements.
- Loudness vs Instrumentalness (-0.33): Instrumental tracks are less loud, aligning with their typically softer, background nature.

Danceability & Tempo

- Danceability vs Tempo (-0.19): Faster tempo doesn't necessarily mean more danceable. In fact, overly fast songs may reduce perceived danceability.
- Danceability vs Liveness (-0.18): Live recordings are slightly less danceable, perhaps due to crowd noise or less polished production.

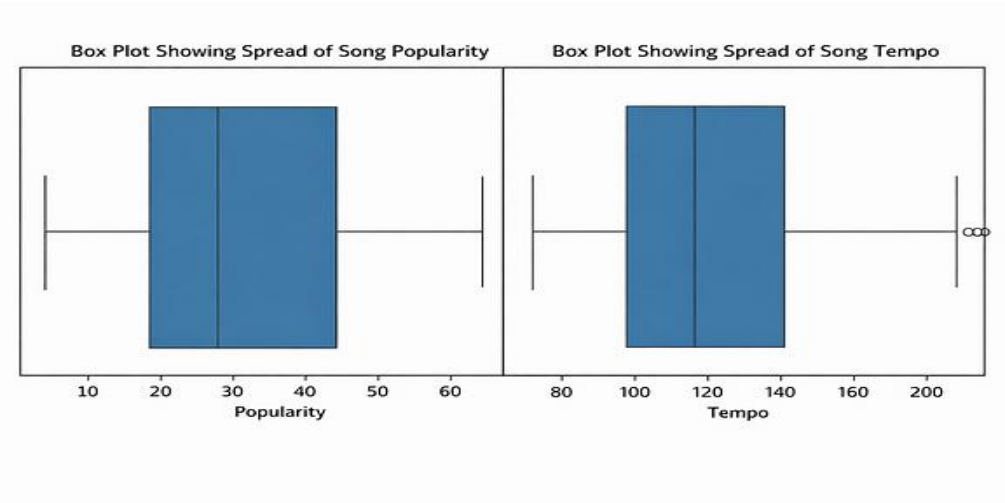
Speechiness

- Speechiness vs Loudness (0.19): Songs with spoken word elements (like rap) tend to be louder.
- Speechiness vs Acousticness (-0.22): Spoken-word-heavy tracks are less acoustic, reflecting their reliance on production.

General Patterns

- Most correlations are weak to moderate, meaning no single feature dominates popularity or musical style.
- The strongest relationships are:
    - Loudness vs Acousticness (-0.53) → clear stylistic divide.
    - Popularity vs Instrumentalness (-0.28) → vocals matter.
    - Popularity vs Acousticness (0.27) → acoustic appeal boosts popularity.

Box Plot:



Distribution Overview:

- Median Popularity: ~30
- Interquartile Range (IQR): 15 to 45
- Whiskers Range: ~5 to 65

Key Insights:

- Most songs fall within moderate popularity levels.
- Very few songs are extremely popular or unpopular.
- The spread suggests popularity is variable, influenced by multiple musical features (like vocals, acousticness, etc.).

Box Plot: Song Tempo:

Distribution Overview:

- Median Tempo: ~130 BPM
- Interquartile Range (IQR): 100 to 140 BPM
- Whiskers Range: ~80 to 190 BPM
- Outliers: ~200 BPM

Key Insights:

- Most songs are in the moderate-to-fast tempo range.
- Tempo is more standardized than popularity, clustering around typical pop/dance BPM.
- Outliers indicate a few songs are unusually fast, but not necessarily more popular.

Business Implications:

1.Music Production Strategy

- Tempo Sweet Spot: Most popular songs cluster around 120–130 BPM. Producers aiming for mainstream success should target this tempo range to align with listener expectations.
- Avoid Extremes: Extremely fast or slow tempos are rare and may appeal to niche audiences. Staying within the interquartile range increases mass appeal.

2. Marketing & Promotion

- Popularity Spread: Since popularity varies widely, marketing efforts can help push moderately popular songs into higher tiers. The box plot shows that many songs hover near the median — a good sign for growth potential.
- Data-Driven Targeting: Understanding where a song falls in the popularity distribution helps tailor promotional strategies (e.g., viral campaigns for mid-tier songs vs. sustaining interest for top-tier hits).

3. Streaming Platform Optimization

- Algorithmic Curation: Platforms can use tempo and popularity data to refine playlists. For example, "Top Pop Hits" might favor songs in the 100–140 BPM range with mid-to-high popularity scores.
- User Retention: Matching listener preferences with tempo trends can improve engagement and reduce skip rates.

Real-Life Meaning

1. Listener Behavior

- People tend to prefer songs with moderate tempo — not too slow, not too fast. This reflects emotional comfort and rhythmic familiarity.
- Popularity isn't just about quality — it's shaped by exposure, genre, and production choices.

2. Artistic Decisions

- Artists can use this data to balance creativity with commercial viability. For example, choosing a tempo around 120 BPM might increase the chance of radio play or playlist inclusion.

3. Cultural Trends

- The clustering of tempo around 120–130 BPM reflects the dominance of pop, dance, and hip-hop genres in recent years.
- Popularity distribution shows that virality **and reach** are key — not every song needs to be a chart-topper to be successful.

6.Conclusion

- This project focused on performing exploratory data analysis on a music dataset to understand the key factors that influence song popularity. The dataset was carefully processed, cleaned, and analysed using Python libraries to ensure accurate and meaningful results. Through systematic analysis, important patterns and relationships within the data were identified.

- A wide range of visualization techniques such as pie charts, bar charts, scatter plots, histograms, box plots, and heatmaps were used to explore the dataset. These visualizations helped in understanding the distribution of features, comparing different categories, and identifying relationships between variables like danceability, tempo, genre, and popularity.

- The analysis showed that song popularity is not dependent on a single feature but rather on a combination of multiple attributes. While factors like genre, danceability, and tempo have some influence, their impact is not very strong individually. The presence of outliers and variations in the dataset also indicates that listener preferences can vary widely.

- Correlation analysis further revealed that most features have weak to moderate relationships with each other, suggesting that predicting popularity is complex. This highlights the importance of considering multiple features together rather than relying on one single factor.

- Overall, this project provides valuable insights into music trends and user preferences. The findings can be useful for artists, producers, and streaming platforms to make informed decisions regarding content creation, promotion, and recommendation systems. It also lays a foundation for future work involving advanced analytics and predictive modelling to better understand and forecast song popularity.

- In addition to the current analysis, this project can be further extended by applying machine learning techniques to build predictive models for song popularity. By using algorithms such as regression, decision trees, or clustering, it is possible to uncover deeper patterns and improve prediction accuracy. Incorporating additional data such as listener demographics, streaming behavior, and social media trends can also enhance the analysis. This would help in developing more personalized music recommendation systems and provide a more comprehensive understanding of what drives a song's success in the real world.

7.Future Scope

1. Machine learning models can be applied to predict song popularity based on multiple audio features and improve accuracy of analysis.
2. The study can be extended by using larger and real-time datasets to get more updated and reliable insights.
3. Additional factors such as artist popularity, number of followers, and album success can be included for deeper analysis.
4. Social media trends and user engagement data can be analysed to understand their impact on song popularity.
5. Advanced recommendation systems can be developed to suggest songs based on user preferences and listening behaviour.
6. A detailed genre-wise analysis can be performed to better understand audience interests in different types of music.
7. Interactive dashboards and advanced visualization tools can be used to present insights more effectively.
8. Deep learning techniques can be applied to capture complex patterns in music data.
9. Regional and cultural preferences can be studied to understand variations in music taste across different locations.
10. Integration with streaming platform data can provide real-time insights and help in making practical business decisions.

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- Seaborn Documentation. Available at: https://seaborn.pydata.org

Appendix

Python Code (Google Colab Notebook)

Google Colab link:
https://colab.research.google.com/drive/1Eow5_Vp1MXOubfx3DEaZkEsuvB5bEwnQ

 # Import Libraries

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns
```

Figure A1: Distribution of Explicit and Non-Explicit Songs

```
plt.figure()

df["explicit"].value_counts().plot(kind="pie", autopct="%1.1f%%")

plt.title("Distribution of Explicit and Non Explicit Songs")

plt.ylabel("")

plt.show()
```

Figure A2: Average Popularity of Top 10 Music Genres

```
plt.figure()

df.groupby("track_genre")["popularity"].mean().sort_values(ascending=False).head(10).plot(
kind="bar")

plt.title("Average Popularity of Top 10 Music Genres")

plt.xlabel("Genre")

plt.ylabel("Average Popularity")

plt.show()
```

Figure A3: Relationship Between Danceability and Song Popularity

```
plt.figure()
plt.scatter(df["danceability"], df["popularity"])
plt.title("Relationship Between Danceability and Song Popularity")
plt.xlabel("Danceability")
plt.ylabel("Popularity")
plt.show()
```

Figure A4: Relationship Between Tempo and Song Popularity

```
plt.figure()
plt.scatter(df["tempo"], df["popularity"])
plt.title("Relationship Between Tempo and Song Popularity")
plt.xlabel("Tempo")
plt.ylabel("Popularity")
plt.show()
```

Figure A5: Distribution of Song Popularity

```
plt.figure()
df["popularity"].hist()
plt.title("Distribution of Song Popularity")
plt.xlabel("Popularity")
plt.ylabel("Number of Songs")
plt.show()
```

Figure A6: Distribution of Song Tempo

```
plt.figure()
df["tempo"].hist()
plt.title("Distribution of Song Tempo")
plt.xlabel("Tempo")
plt.ylabel("Number of Songs")
plt.show()
```

Figure A7: Trend of Song Tempo Across Dataset

```
plt.figure()
df_sorted = df.sort_values("tempo").reset_index()
plt.plot(df_sorted["index"], df_sorted["tempo"])
plt.title("Trend of Song Tempo Across Dataset")
plt.xlabel("Index")
plt.ylabel("Tempo")
plt.show()
```

Figure A8: Trend of Song Popularity Across Dataset

```
plt.figure()
df_sorted = df.sort_values("popularity").reset_index()
plt.plot(df_sorted["index"], df_sorted["popularity"])
plt.title("Trend of Song Popularity Across Dataset")
plt.xlabel("Index")
```

```
plt.ylabel("Popularity")

plt.show()
```

Figure A9: Correlation Heatmap of Music Features

```
plt.figure()

corr =
df[["popularity","duration_ms","danceability","loudness","speechiness","acousticness","instr
umentalness","liveness","tempo"]].corr()

sns.heatmap(corr, annot=True)

plt.title("Correlation Heatmap of Music Features")

plt.show()
```

Figure A10: Box Plot Showing Spread of Song Popularity

```
plt.figure()

sns.boxplot(x=df["popularity"])

plt.title("Box Plot Showing Spread of Song Popularity")

plt.xlabel("Popularity")

plt.show()
```

Figure A11: Box Plot Showing Spread of Song Tempo

```
plt.figure()

sns.boxplot(x=df["tempo"])

plt.title("Box Plot Showing Spread of Song Tempo")

plt.xlabel("Tempo")

plt.show()
```

Figure A12: Average Popularity of Explicit vs. Non-Explicit Songs

```
plt.figure()

sns.barplot(x="explicit", y="popularity", data=df)

plt.title("Average Popularity of Explicit vs Non Explicit Songs")

plt.xlabel("Explicit")

plt.ylabel("Average Popularity")

plt.show()
```